Advanced Computer-Aided VLSI System Design
Lab: Timing Signoff (PrimeTime)

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Objectives

In this lab, you will learn how to use PrimeTime to perform post-route STA and timing/power/drv ECO with DMSA.

Data Preparation

1. Create a work directory and copy the lab files from
   `/share1/home/acvsd-ta11/lab6.tar`
2. Decompress the file and check if you have these files:

Filename	Description
CHIP.def	Physical information for CHIP
CHIP_rc_best.spef	Post-route RC file for CHIP (RC best corner)
CHIP_rc_worst.spef	Post-route RC file for CHIP (RC worst corner)
CHIP_pr.v	Post-route Verilog file for CHIP
common.tcl	Common script for PT DMSA
specific_max.tcl	Function mode constraints for CHIP (for setup check)
specific_min.tcl	Scan mode constraints for CHIP (for setup check)
pt.tcl	Main script for PrimeTime DMSA ECO
Library/*	Physical library for CHIP (lef/tlef)

3. Check the contents of these files.

Invoke PrimeTime

To invoke PrimeTime, first include the following C shell script

```
% cb
```

And then, you can do either one

`% pt_shell` (command mode)

`% pt_shell -gui` (GUI mode)

In this Lab, we will use the command mode. You are welcome to try the GUI mode. In the GUI mode, you can type in your commands in the command line, located at the bottom of the window.

STEP 1: Check the common.tcl (No need to do anything)

```
##############################
# ECO option
##############################

set_eco_options -physical_tech_lib_path library/N16ADFP_APR_Innovus_11M.10a.tlef \
                -physical_lib_path { \
                      library/N16ADFP_StdCell.lef library/N16ADFP_StdIO.lef \
                      library/N16ADFP_BondPad.lef library/rf_2p_hse.lef} \
                 -physical_design_path CHIP.def
```

Explanation: PrimeTime DMSA will take the physical information to perform ECO using technology lef, physical frame of IP lef and the path, pin, cell orientation and routes from def file.

```
########################################
# OCV Setting
########################################

set_timing_derate -early 0.9
set_timing_derate -late 1.1
# ===This part is used if noise model, pocvm side file, ccs_lvf are applicable====#
# set_ocvm pocv.table
# set_ocvm pocv.distance
# set_app_var timing_pocvm_enable_analysis true
# set_app_var timing_enable_constraint_variation true
# set_app_var timing_enable_slew_variation true
# set_app_var timing_pocvm_corner_sigma 3.0
# set_app_var timing_pocvm_report_sigma 3.0
# set_app_var timing_pocvm_max_transition_sigma 3.0
# set_app_var si_enable_analysis true
# report_ocvm -type pocvm
# report_timing -variation -crosstalk_delta
```

Explanation: In timing sign-off, timing derate and variation setting is very important. However, ADFP doesn't include the LVF lib file or POCV side file. The crosstalk and SI analysis are also not applicable because of lacking of noise models on liberty file. In this lab we will only use traditional global ocv (set_timing_derate). But we strongly

recommend that you know the commented part, because in industry, you will use these advanced commands a lot.

STEP 2: Check the specific_max/min.tcl (No need to do anything)

```
# In specific_max.tcl
set_operating_conditions -analysis_type on_chip_variation -library \
    N16ADFP_StdCellss0p72v125c_ccs
read_parasitics -keep_capacitive_coupling -format SPEF ./CHIP_rc_worst.spef

# In specific_min.tcl
set_operating_conditions -analysis_type on_chip_variation -library \
    N16ADFP_StdCellff0p88vm40c_ccs
read_parasitics -keep_capacitive_coupling -format SPEF ./CHIP_rc_best.spef
```

Explanation: In read_parasitics, the RC corner is read into the design for RC back-annotation. We also strongly recommend that you can take a look into the spef files, it can be extracted from RC Extraction tool like: Calibre RCX, StarRC or Quantus. The analysis type should be set to on_chip_variation to include ocv timing analysis.

STEP 3: Source the related libraries

```
history keep 100
alias h history

##############################################
# set libraries
##############################################
set search_path " \
    /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/ \
    /share1/tech/ADFP/Executable_Package/Collaterals/IP/sram/N16ADFP_SRAM/NLDM/ \
    /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/NLDM/ \
    /share1/tech/ADFP/Lab_adfp/library/memory/ \
    $search_path .\
"

set target_library " \
    N16ADFP_StdCellss0p72v125c_ccs.db \
    N16ADFP_StdCellff0p88vm40c_ccs.db \
    N16ADFP_SRAM_ss0p72v0p72v125c_100a.db \
    N16ADFP_SRAM_ff0p88v0p88vm40c_100a.db \
    N16ADFP_StdIOss0p72v1p62v125c.db \
    N16ADFP_StdIOff0p88v1p98vm40c.db \
    rf_2p_hse_nldm_ss_0p72v_0p72v_125c_syn.db \
    rf_2p_hse_nldm_ff_0p88v_0p88v_m40c_syn.db \
"

set link_library "* $target_library"
```

STEP 4: Setting the DMSA Configuration

Execute the seach_path and library setting part first in pt.tcl and run the below command:

```
##########################################
# DMSA Setting
##########################################
set_app_var multi_scenario_enable_analysis true
create_scenario -name func_max -common_data ./common.tcl -specific_data ./specific_max.tcl
create_scenario -name scan_min -common_data ./common.tcl -specific_data ./specific_min.tcl
set_host_options -name my_option -max_cores 4 -num_processes 4
report_hosts > host_options.log
start_host
current_session {func_max scan_min}
```

This script enables DMSA, allowing parallel processing of multiple timing scenarios such as functional and scan modes. It defines shared and scenario-specific data files, configures host resource usage (cores and processes), initializes the host environment, and activates the desired scenarios for concurrent analysis. This setup helps improve runtime efficiency and ensures consistent analysis across multiple corners or modes.

```
****************************************
Report : host_usage
Version: P-2019.03-SP5-1
Date   : Sun May 11 00:12:31 2025
****************************************

  Options Name          Host Name    Num Processes           Protocol
  ------------------------------------------------------------------
  my_option             localhost    4                       auto

  Usage limits (cores)

  Options Name     #                                       Effective
  ------------------------------------------------------------------
  (local process)  -                                       4
  my_option                                                -
  ------------------------------------------------------------------
  Total                                                    4
```

If you are successful with setting, you can see the host_option.log as above.

STEP 5: Pre-ECO Timing Report

Execute the following script to get the postroute STA timing report in DSMA mode.

```
########################################
# Pre ECO report
########################################

remote_execute {
    report_global_timing
    set_app_var timing_save_pin_arrival_and_required true
    set_app_var power_enable_analysis true
    update_power
}

set_app_var timing_save_pin_arrival_and_required true
set_app_var power_enable_analysis true
report_power > pre_eco.power
report_global_timing > pre_eco.timing
report_timing -delay_type min -path_type full_clock_expanded -significant_digits 4 \
    -capacitance -transition_time -derate -nosplit > pre_eco_detail_min_path.log
report_timing -delay_type max -path_type full_clock_expanded -significant_digits 4 \
    -capacitance -transition_time -derate -nosplit > pre_eco_detail_max_path.log
```

This script generates reference reports before applying any ECO modifications. It first enables power and timing analysis, saves pin arrival/required timing data, and updates power estimates. The script then outputs reports including global timing, power, and detailed min/max timing paths (with capacitance, transition time, and derate information). These serve as a baseline for later comparison to post-ECO results.

After applying the timing derate, you will find negative hold slack in pre_eco.timing and pre_eco_detail_min_path.log.

```
Startpoint: u_IDCT/tposemem_Count_reg[0]
            (rising edge-triggered flip-flop clocked by CLK1)
Endpoint: u_IDCT/tposemem_Bisted_RF_2P_ADV64x16_RF_2P_ADV64x16_u0_i_rf_2p_SRAM_i1
            (rising edge-triggered flip-flop clocked by CLK1)
Last common pin: u_IDCT/CTS_ccl_a_buf_00693/Z
Path Group: CLK1
Path Type: min
Scenario: scan_min

Point                                                
```

STEP 6: ECO leakage power and Timing

Execute the following script to perform ECO in DSMA mode and write the ECO changes.

```
########################################
# ECO leakage power & Timing
########################################

fix_eco_power -power_mode leakage -leakage_scenario {func_max} -verbose
write_changes -format text -output eco_leakage_changes.txt
write_changes -format icc2tcl -output icc2_dmsa_eco_leakage_script.tcl
write_changes -format eco -output pt_dmsa_eco_leakage.dat

fix_eco_timing -type hold -buffer_list {BUFFD1BWP16P90 BUFFD2BWP16P90 BUFFD4BWP16P90 \
    BUFFD1BWP16P90LVT BUFFD2BWP16P90LVT BUFFD4BWP16P90LVT} -verbose
write_changes -format text -output eco_hold_changes.txt
write_changes -format icc2tcl -output icc2_dmsa_eco_hold_script.tcl
write_changes -format eco -output pt_dmsa_eco_hold.dat
```

This script performs ECO optimization for leakage power and hold timing. It first applies power ECO under the func_max scenario to reduce leakage, then generates output files in text, TCL, and ECO formats for review and downstream usage. Subsequently, it fixes hold timing violations using a specified list of buffers (including both regular and LVT types), and similarly exports the resulting changes in multiple formats. This process ensures the design meets both power and timing requirements with consistent documentation.

STEP 7: Write out Post ECO Timing Report

Execute the following script to get the Post ECO timing reports in DSMA mode.

```
########################################
# Post ECO report
########################################

report_power > post_eco.power
report_global_timing > post_eco.timing
report_timing -delay_type min -path_type full_clock_expanded -significant_digits 4 \
    -capacitance -transition_time -derate -nosplit > post_eco_detail_min_path.log
report_timing -delay_type max -path_type full_clock_expanded -significant_digits 4 \
    -capacitance -transition_time -derate -nosplit > post_eco_detail_max_path.log
```

This script generates power and timing reports after ECO changes have been applied. It outputs updated power (post_eco.power) and global timing (post_eco.timing) data, along with detailed min and max path timing reports. These reports include capacitance, transition time, and derate information, enabling comparison with pre-ECO data to verify that timing and power goals are preserved or improved after the fixes. After ECO you will find there will not have any hold violation now.

```
Start of Master/Slave Task Processing
-------------------------------------
Started    : Command execution in scenario 'func_max'
Started    : Command execution in scenario 'scan_min'
Succeeded  : Command execution in scenario 'func_max'
Succeeded  : Command execution in scenario 'scan_min'
---------------------------------
End of Master/Slave Task Processing

No setup violations found.

No hold violations found.
1
```

```
Startpoint: u_DCT/tposemem_Count_reg[3]
            (rising edge-triggered flip-flop clocked by CLK1)
Endpoint: u_DCT/tposemem_Bisted_RF_2P_ADV64x16_RF_2P_ADV64x16_u0_i_rf_2p_SRAM_i0
            (rising edge-triggered flip-flop clocked by CLK1)
Last common pin: CTS_ccl_a_buf_00718/Z
Path Group: CLK1
Path Type: min
Scenario: scan_min

Point                                                                                                                  Cap    Trans    Derate    Incr       Path
-------------------------------------------------------------------------------------------------------------------------------------------------------------------
clock CLK1 (rise edge)                                                                                                                          0.0000     0.0000
clock source latency                                                                                                                            0.0000     0.0000
CLK (in)                                                                                                             0.0795   0.0080            0.0072     0.0072 r
ipad_CLK/PAD (PDCDG_V)                                                                                                        0.0080   0.9000   0.0000     0.0072 r
ipad_CLK/C (PDCDG_V)                                                                                                 0.0037   0.0104   0.9000   0.1321     0.1393 r
CTS_cpc_drv_buf_00752/I (CKBD12BWP16P90LVT)                                                                                   0.0104   0.9000   0.0000     0.1393 r
CTS_cpc_drv_buf_00752/Z (CKBD12BWP16P90LVT)                                                                          0.0045   0.0049   0.9000   0.0112     0.1505 r
CTS_ccl_buf_00734/I (CKBD16BWP16P90LVT)                                                                                       0.0049   0.9000   0.0000     0.1505 r
CTS_ccl_buf_00734/Z (CKBD16BWP16P90LVT)                                                                              0.0146   0.0068   0.9000   0.0119     0.1624 r
CTS_ccl_buf_00726/I (CKBD8BWP16P90LVT)                                                                                        0.0068   0.9000   0.0000     0.1624 r
CTS_ccl_buf_00726/Z (CKBD8BWP16P90LVT)                                                                               0.0047   0.0052   0.9000   0.0105     0.1729 r
CTS_ccl_a_buf_00718/I (CKBD8BWP16P90LVT)                                                                                      0.0052   0.9000   0.0000     0.1729 r
CTS_ccl_a_buf_00718/Z (CKBD8BWP16P90LVT)                                                                             0.0082   0.0066   0.9000   0.0112     0.1841 r
CTS_ccl_a_buf_00705/I (CKBD4BWP16P90LVT)                                                                                      0.0066   0.9000   0.0000     0.1841 r
CTS_ccl_a_buf_00705/Z (CKBD4BWP16P90LVT)                                                                             0.0099   0.0103   0.9000   0.0126     0.1967 r
u_DCT/RC_CG_DECLONE_HIER_INST3736/RC_CGIC_INST/CP (CKLNQD2BWP16P90LVT)                                                        0.0103   0.9000   0.0000     0.1967 r
u_DCT/RC_CG_DECLONE_HIER_INST3736/RC_CGIC_INST/Q (CKLNQD2BWP16P90LVT)                                                0.0142   0.0274   0.9000   0.0304     0.2271 r
u_DCT/tposemem_Count_reg[3]/CP (SDFQND1BWP20P90)                                                                              0.0274   0.9000   0.0000     0.2271 r
u_DCT/tposemem_Count_reg[3]/QN (SDFQND1BWP20P90)                                                                     0.0011   0.0154   0.9000   0.0608     0.2879 r
u_DCT/FE_PHC441_tposemem_Count_3/I (CKBD1BWP20P90)                                                                            0.0154   0.9000   0.0000     0.2879 r
u_DCT/FE_PHC441_tposemem_Count_3/Z (CKBD1BWP20P90)                                                                   0.0069   0.0333   0.9000   0.0288     0.3167 r
u_DCT/U_PTECO_HOLD_BUF6/I (DEL050D1BWP16P90LVT)                                                                               0.0333   0.9000   0.0000     0.3167 r
u_DCT/U_PTECO_HOLD_BUF6/Z (DEL050D1BWP16P90LVT)                                                                      0.0011   0.0060   0.9000   0.0518 +   0.3686 r
u_DCT/FE_PHC471_tposemem_Count_3/I (BUFFD1BWP20P90)                                                                           0.0060   0.9000   0.0000 +   0.3686 r
u_DCT/FE_PHC471_tposemem_Count_3/Z (BUFFD1BWP20P90)                                                                  0.0006   0.0060   0.9000   0.0119     0.3804 r
u_DCT/g3684/A2 (AOI22D1BWP20P90)                                                                                              0.0060   0.9000   0.0000     0.3804 r
u_DCT/g3684/ZN (AOI22D1BWP20P90)                                                                                     0.0011   0.0131   0.9000   0.0116     0.3920 f
u_DCT/FE_PHC469_n_144/I (BUFFD1BWP20P90)                                                                                      0.0131   0.9000   0.0000     0.3920 f
u_DCT/FE_PHC469_n_144/Z (BUFFD1BWP20P90)                                                                             0.0006   0.0060   0.9000   0.0138     0.4058 f
u_DCT/g3681/A1 (ND2D1BWP20P90)                                                                                                0.0060   0.9000   0.0000     0.4058 f
u_DCT/g3681/ZN (ND2D1BWP20P90)                                                                                       0.0103   0.0481   0.9000   0.0284     0.4342 r
u_DCT/tposemem_Bisted_RF_2P_ADV64x16_RF_2P_ADV64x16_u0_i_rf_2p_SRAM_i0/AA[3] (rf_2p_hse)                                      0.0481   0.9000   0.0000     0.4342 r
data arrival time                                                                                                                                         0.4342

clock CLK1 (rise edge)                                                                                                                          0.0000     0.0000
clock source latency                                                                                                                            0.0000     0.0000
CLK (in)                                                                                                             0.0795   0.0080            0.0087     0.0087 r
ipad_CLK/PAD (PDCDG_V)                                                                                                        0.0080   1.1000   0.0000     0.0087 r
ipad_CLK/C (PDCDG_V)                                                                                                 0.0037   0.0104   1.1000   0.1615     0.1703 r
CTS_cpc_drv_buf_00752/I (CKBD12BWP16P90LVT)                                                                                   0.0104   1.1000   0.0000     0.1703 r
CTS_cpc_drv_buf_00752/Z (CKBD12BWP16P90LVT)                                                                          0.0045   0.0049   1.1000   0.0137     0.1840 r
CTS_ccl_buf_00734/I (CKBD16BWP16P90LVT)                                                                                       0.0049   1.1000   0.0000     0.1840 r
CTS_ccl_buf_00734/Z (CKBD16BWP16P90LVT)                                                                              0.0154   0.0069   1.1000   0.0147     0.1986 r
CTS_ccl_buf_00726/I (CKBD8BWP16P90LVT)                                                                                        0.0069   1.1000   0.0000     0.1986 r
CTS_ccl_buf_00726/Z (CKBD8BWP16P90LVT)                                                                               0.0047   0.0053   1.1000   0.0129     0.2115 r
CTS_ccl_a_buf_00718/I (CKBD8BWP16P90LVT)                                                                                      0.0053   1.1000   0.0000     0.2115 r
CTS_ccl_a_buf_00718/Z (CKBD8BWP16P90LVT)                                                                             0.0084   0.0066   1.1000   0.0137     0.2253 r
CTS_ccl_a_buf_00703/I (CKBD12BWP16P90LVT)                                                                                     0.0066   1.1000   0.0000     0.2253 r
CTS_ccl_a_buf_00703/Z (CKBD12BWP16P90LVT)                                                                            0.0169   0.0081   1.1000   0.0155     0.2408 r
u_DCT/CTS_ccl_a_buf_00660/I (CKBD8BWP16P90LVT)                                                                                0.0081   1.1000   0.0000     0.2408 r
u_DCT/CTS_ccl_a_buf_00660/Z (CKBD8BWP16P90LVT)                                                                       0.0609   0.0278   1.1000   0.0286     0.2693 r
u_DCT/tposemem_Bisted_RF_2P_ADV64x16_RF_2P_ADV64x16_u0_i_rf_2p_SRAM_i0/CLKA (rf_2p_hse)                                       0.0278   1.1000   0.0000     0.2693 r
clock reconvergence pessimism                                                                                                                  -0.0412     0.2282
clock uncertainty                                                                                                                               0.0200     0.2482
library hold time                                                                                                                    1.0000     0.1860     0.4341
data required time                                                                                                                                        0.4341
-------------------------------------------------------------------------------------------------------------------------------------------------------------------
data required time                                                                                                                                        0.4341
data arrival time                                                                                                                                        -0.4342
-------------------------------------------------------------------------------------------------------------------------------------------------------------------
slack (MET)                                                                                                                                               0.0001
```

Checkpoints

Please check with TAs before leaving this lab to make sure the following goals are accomplished and to get credits.

1. Show your pre and post_eco_detail_min_path.log.
2. Show the power before and after ECO. How much does the leakage power be reduced.
3. Answer the questions : if the DRV and SI are violation how to eco the violation in PT-DMSA? (Check Handout)